What Do Time Series Foundation Models Actually Learn? A Geometric Perspective

Abstract

Time-series foundation models have achieved strong empirical performance, yet the nature of the representations they learn remains poorly understood. In this work, we conduct a systematic analysis of the internal representations of a large pretrained time-series model, contrasting them with those of an identical architecture under random initialization. Across four widely used benchmarks, we find that pretraining consistently induces highly structured representation spaces, in contrast to the diffuse and uninformative geometry observed in randomly initialized counterparts.

Leveraging this structure, we introduce a simple, label-free reliability indicator based on representation-space proximity: inputs that are distant from their nearest neighbors—i.e., “unfamiliar” to the model—exhibit significantly higher forecast error. This provides a practical mechanism for uncertainty estimation without requiring additional supervision or architectural modifications.

We further probe the semantic content of the learned representations and show that they robustly encode recurring temporal patterns and variability, while remaining largely insensitive to trend direction. Finally, we evaluate transfer under distribution shift using out-of-distribution equity data. While structural and pattern-based features generalize effectively, the proposed reliability signal degrades, suggesting a decoupling between representation quality and uncertainty estimation under shift. Together, our findings offer new insights into what time-series foundation models learn and how their representations can be leveraged—and limited—in downstream applications.

1. Introduction

The time series community is at an inflection point. Foundation models—TimesFM **?**, Chronos **?**, MOMENT **?**, Moirai **?**, Lag-Llama **?**—pretrained on large, heterogeneous temporal corpora, promise zero-shot transfer across energy, climate, finance, and traffic. The promise has generated intense excitement. It has also generated intense scepticism.

? showed that a single-layer linear model matches or outperforms deep Transformer forecasters on five standard benchmarks—raising the uncomfortable question of whether architectural complexity contributes anything at all to forecasting quality. Subsequent work has found that simple patch-based supervised models can close much of the zero-shot performance gap when trained on sufficient data **?**, and that TSFM zero-shot gains vary substantially across evaluation protocols, preprocessing choices, and prediction horizons. The community has spent two years debating *whether* TSFMs outperform baselines.

Yet no one has systematically examined what structure TSFMs actually store in their representations.

This is the question we address. Performance metrics measure model outputs; they reveal nothing about the internal organisation of learned representations. A TSFM that matches a linear model on ETTh1 MAE might encode a geometrically rich, transferable representation of temporal dynamics—or it might encode little beyond distributional statistics of its pretraining corpus. These two scenarios have profoundly different implications for when to trust a zero-shot forecast, how to fine-tune TSFMs for new domains, and whether the scale of pretraining is justified by what is actually learned.

The same debate about performance has been resolved in other domains by asking a deeper question: not *what* a model predicts, but *how* its internal representations are shaped. **?** showed that image classifiers compress representations onto low-dimensional manifolds in terminal layers, with the degree of compression tracking generalisation quality—revealing structure invisible to accuracy metrics alone. **?** demonstrated that BERT organises embeddings into geometrically separable syntactic and semantic subspaces, providing the first structural account of what language pretraining encodes. No such analysis exists for time series models.

We close this gap. Using the **Latent Geometry Framework (LGF)**—a diagnostic suite that operates on frozen TSFM weights without retraining or label access—we conduct the first systematic analysis of the shape and structure of TSFM representation spaces. Our main findings are:

- **TSFMs develop richer temporal structure in their internal representations.** Across four benchmarks, pretrained TSFM embeddings move through space in more varied, structured paths than randomly-initialised controls, yet remain about $2.25\times$ more compact than supervised Transformer baselines on Electricity under a unified evaluation protocol. This structure is absent in randomly-initialised controls, confirming it is learned from diverse temporal data.
- **The model’s embedding space encodes forecast reliabil-

ity. For pretrained models, windows that sit far from their neighbours in embedding space tend to produce larger, less consistent forecast errors—and this local-proximity signal (PDS) is far stronger than anything captured by global pairwise distances (FSS) or by the same measure applied to randomly-initialised models. This yields a label-free, training-free reliability signal for zero-shot deployment.

- **Basic temporal patterns are directly readable from frozen embeddings.** A simple linear classifier trained on frozen TSFM embeddings recovers seasonality and volatility with significantly above-chance accuracy (82.6% and 77.6% on Electricity), outperforming randomly-initialised controls by up to 18 percentage points. This is the first direct account of what temporal patterns are stored in pretrained TSFM representations.

Our analysis covers TimesFM 2.0 (500M-parameter decoder-only) evaluated against three supervised baselines and a random-weight control on four standard benchmarks. All findings are observational; we characterise geometric structure without claiming it causes forecasting quality.

2. Preliminaries and Related Work

2.1. Time Series Foundation Models

The past two years have produced a generation of models pretrained on large, heterogeneous time series corpora with the goal of zero-shot or few-shot transfer **??**. These models differ along two design axes: how input data is divided into tokens and what prediction target guides training.

Patch-based encoder-decoder models divide the context window into fixed-length patches and learn representations via masked modelling. MOMENT **?** applies this to sub-series segments, supporting forecasting, classification, and anomaly detection. Moirai **?** extends it with a multi-patch-size architecture that accommodates variable-frequency series.

Decoder-only autoregressive models treat forecasting as next-patch prediction. TimesFM **?** trains a 500M-parameter model on a corpus spanning Google Trends, Wikipedia pageviews, and synthetic series. Timer **?** follows the same autoregressive design at smaller scale.

Language-model-inspired models convert continuous series values into discrete tokens—treating numerical forecasting as a text-prediction problem—and fine-tune sequence-to-sequence architectures. Chronos **?** builds on T5 **?**, casting forecasting as a sequence-to-sequence language task. Lag-Llama **?** adapts LLaMA by appending lagged copies of the input series as additional context features. Concurrent work has also explored prompting general-purpose LLMs directly for zero-shot forecasting **?**, with mixed results that further illustrate the gap between raw performance and what the model has actually learned to represent.

2.2. The Performance Debate: Promise vs. Reality

The rapid proliferation of TSFMs has been accompanied by an equally rapid accumulation of contradictory evidence about their practical utility. Understanding this debate is essential context for our geometric analysis.

The sceptical challenge. ? delivered the most influential critique: a one-layer linear model (DLinear) matches or outperforms every major Transformer-based forecaster of its era on five standard long-horizon benchmarks, including ETT and Weather. The implication is stark: if a model with no representational capacity matches a deeply stacked Transformer, either the Transformer is learning something irrelevant to the task, or the benchmarks are not measuring what matters. This result reframed the research agenda from "which Transformer is best" to "does Transformer complexity help at all".

The zero-shot argument. Proponents of TSFMs respond that the comparison is unfair: DLinear is trained per-dataset, while TSFMs operate zero-shot. The Chronos **?** and TimesFM **?** evaluations both show zero-shot performance competitive with, and sometimes exceeding, supervised per-dataset models—a capability that linear models cannot replicate by definition. Moirai **?** further demonstrates that a single foundation model can handle variable frequencies and multivariate series within a unified architecture.

The unresolved tension. Despite this back-and-forth, several uncomfortable facts remain. First, the zero-shot superiority of TSFMs is not universal: on standard long-horizon benchmarks with sufficient training data, supervised patch-based models like PatchTST **?** remain highly competitive at a fraction of the parameter count. Second, TSFM evaluations are not standardised: reported results vary significantly across evaluation protocols, preprocessing choices, and prediction horizons **?**, making apples-to-apples comparison difficult. Third, and most importantly for our work: *the debate is conducted entirely at the output level*. Neither proponents nor sceptics have systematically examined what these models actually store in their internal representations.

The missing question. A model that achieves competitive zero-shot MAE on ETTh1 could do so by (a) building a rich, transferable internal representation of temporal patterns, or (b) memorising statistical properties of the pretraining data that happen to generalise. These two mechanisms have radically different implications for when TSFMs generalise, when they fail, and how they should be used. Performance metrics cannot distinguish them. Geometric analysis of representations can. This is the contribution of the present paper.

2.3. Geometric Analysis of Neural Representations

The internal geometry of learned representations has been studied systematically in vision and language, providing the methodological toolkit we adapt to time series.

Intrinsic dimensionality. ? showed that deep image classifiers exhibit a characteristic layer-wise intrinsic dimensionality (ID) profile: expansion in early layers, compression to a low-dimensional manifold in terminal layers, with the compression ratio correlating with generalisation. **?** extended this to language models, showing that low intrinsic dimensionality of the fine-tuning landscape explains why large pretrained models adapt efficiently with few parameters. The TwoNN estimator **?** provides a non-parametric method for estimating this dimension from nearest-neighbour distance ratios—part of the geometric toolkit we adapt to time series representations.

Representational geometry in language models. **?** demonstrated that BERT organises contextual embeddings into geometrically separable clusters corresponding to syntactic and semantic categories—a structural finding invisible to accuracy benchmarks alone. **?** used SVCCA to show that lower layers stabilise early in training while upper layers continue evolving, showing how representations form layer by layer and motivating the analysis of what learned representations store.

The time series gap. No equivalent geometric characterisation exists for time series models. We do not know whether TSFMs organise their embedding spaces around temporal patterns, whether their internal structure is stable under small input changes, or whether proximity in embedding space predicts forecast reliability. These are the questions LGF is designed to answer.

2.4. Explainability for Time Series

Existing XAI approaches for time series assign importance scores to input timesteps—via LIME, SHAP, or integrated gradients—or measure model sensitivity to input perturbations **??**. Both classes operate at the input level and explain individual predictions. Neither provides any account of what a model has learned to encode in its representation space.

Linear probing **?** bridges this gap: it tests whether a target property can be read out of frozen intermediate representations using a linear classifier, revealing what information is stored without the confounds of end-to-end training. In NLP, probing revealed that BERT encodes syntactic structure in intermediate layers and semantic structure in upper layers **?**. We apply this approach to provide the first direct account of what temporal information is stored in pretrained TSFM representations.

3. Latent Geometry Framework

When a model processes a time series window, it maps it to a point in a high-dimensional vector space. LGF treats the collection of all such points as a geometric object—a *manifold*—and asks: is this manifold shaped by what was learned, or merely by the architecture? Does it organise itself around temporal concepts? Does its local structure predict how reliably the model will forecast?

We formalise this intuition as a four-stage diagnostic pipeline. Given windows drawn from a target time series and paired forecast errors, LGF: (1) measures how embeddings move as the time series progresses (*Latent Trajectory Complexity*, LTC); (2) measures how well embeddings cluster by temporal concept (LCS), how stable they are under small input changes (TSI), and whether that cluster structure survives perturbation (GSP); (3) tests whether being far from neighbouring embeddings predicts larger forecast error (PDS), compared to global pairwise distances (FSS); and (4) tests whether basic temporal patterns can be read out of frozen embeddings with a simple linear classifier (*Class Separation Score*, CSS). All stages require only frozen inference—no fine-tuning, gradient access, or label supervision. The full pipeline is summarised in Algorithm 1.

3.1. Problem Setup

Let $f_\theta : \mathbb{R}^L \to \mathbb{R}^D$ denote the mapping from a context window $w \in \mathbb{R}^L$ to a fixed-length embedding vector $z = f_\theta(w) \in \mathbb{R}^D$ produced by a frozen TSFM with $\mathcal{L}$ transformer layers. Sequence embeddings are obtained by mean-pooling the final-layer token representations over $T = L/P$ patch tokens: $z = \frac{1}{T}\sum_{t=1}^{T} f_\theta^{\mathcal{L}}(w)_t$.

Given a dataset $\mathcal{W} = \{(w_i, s_i)\}_{i=1}^N$ of windows with start times s_i and forecast errors $\mathcal{E} = \{e_i\}_{i=1}^N$ at a fixed horizon, let $\mathcal{Z} = \{z_i\}_{i=1}^N$ denote the set of final-layer embeddings. The *Temporal Representation Manifold* (TRM) is the image of the encoder over the support of windows drawn from the target domain:

$$\mathcal{M}_\theta = f_\theta(\mathrm{supp}(\mathcal{P})) \subset \mathbb{R}^D, \tag{1}$$

approximated in practice by $\mathcal{Z}$. Whether $\mathcal{M}_\theta$ has rich temporal structure, stable local geometry, and meaningful cluster organisation are empirical questions that LGF is designed to answer.

3.2. Latent Trajectory Complexity (LTC)

The temporal ordering of windows provides a natural basis for measuring how TSFM embeddings evolve as the time series progresses. Let σ be the permutation sorting windows by start time s_i, and $\delta_i = z_{\sigma(i+1)} - z_{\sigma(i)}$ the embedding step between temporally adjacent windows. We define two complementary quantities:

$$\begin{aligned} \mathrm{LTC}_{\mathrm{step}} &= \frac{1}{N-1}\sum_{i=1}^{N-1} \|\delta_i\|_2, \\ \mathrm{LTC}_{\mathrm{dir}} &= \frac{1}{N-2}\sum_{i=1}^{N-2} \frac{\delta_i \cdot \delta_{i+1}}{\|\delta_i\|\,\|\delta_{i+1}\|}. \end{aligned} \tag{2}$$

$\mathrm{LTC}_{\mathrm{step}}$ is the mean Euclidean step size: how much the embedding moves as the underlying time series advances. $\mathrm{LTC}_{\mathrm{dir}}$ is directional consistency: whether consecutive steps tend to point in the same direction ($\mathrm{LTC}_{\mathrm{dir}}>0$) or change direction unpredictably ($\mathrm{LTC}_{\mathrm{dir}}\approx 0$). A model with richer temporal representations will exhibit larger $\mathrm{LTC}_{\mathrm{step}}$, reflecting structured variation aligned with temporal progression rather than random scatter. Comparing pretrained vs. randomly-initialised models isolates the learned contribution: any significant gap in $\mathrm{LTC}_{\mathrm{step}}$ establishes that richer trajectory structure is a product of pretraining, not architecture.

3.3. Manifold Regularity Metrics

We quantify three complementary aspects of TRM regularity at the terminal layer. LCS measures semantic cluster organisation; TSI measures embedding stability under input corruption independently of cluster structure (a model could be stable yet poorly organised); GSP measures whether perturbation degrades cluster organisation specifically, capturing the intersection of the two. These metrics apply standard geometric tools—silhouette scoring, cosine similarity—to a domain not previously examined this way.

Semantic labels. For each window w_i we derive a trend label $y_i \in \{\mathrm{up}, \mathrm{flat}, \mathrm{down}\}$ from STL decomposition **?**: extract the trend component, compute the OLS slope $\hat{\beta}_i$ over the trend, and assign

$$y_i = \begin{cases} \text{up} & \hat{\beta}_i > \delta \cdot \sigma_{w_i} \\ \text{down} & \hat{\beta}_i < -\delta \cdot \sigma_{w_i} \\ \text{flat} & \text{otherwise,} \end{cases} \tag{3}$$

where $\sigma_{w_i} = \mathrm{std}(w_i)$ and $\delta = 0.05$. Labels are derived entirely from window values, independently of any model. The STL seasonal period is set to the dataset-native frequency (hourly: 24, 10-min: 144, 5-min: 288); sensitivity to $\delta \in \{0.02, 0.05, 0.10\}$ is assessed in Section 4.7.

Latent Cohesion Score (LCS). LCS measures whether embeddings from windows sharing the same trend label cluster together:

$$\mathrm{LCS}(\mathcal{Z}, \mathbf{y}) = \frac{1}{N} \sum_{i=1}^{N} \frac{b(i) - a(i)}{\max\big(a(i),\, b(i)\big)}, \tag{4}$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ the mean nearest-cluster distance for z_i under labels $\mathbf{y}$. LCS $\in [-1, 1]$; values near $+1$ indicate tight, well-separated clusters. In high-dimensional embedding spaces, silhouette-based separation is a conservative measure; linear probing (CSS, §3.5) is more sensitive and serves as the primary cluster-organisation diagnostic.

Temporal Stability Index (TSI). TSI measures the cosine similarity between clean and perturbed embeddings:

$$\mathrm{TSI}_{\mathcal{P}} = \frac{1}{N} \sum_{i=1}^{N} \frac{z_i \cdot \tilde{z}_i}{\|z_i\| \, \|\tilde{z}_i\|}, \quad \tilde{z}_i = f_\theta\big(\mathcal{P}(w_i)\big), \tag{5}$$

evaluated over three types (σ=0.02 noise, p=0.05 dropout, k=3 time-warp). TSI $\in [-1, 1]$; values near $+1$ indicate a stable TRM under realistic input corruption.

Generalised Stability under Perturbation (GSP). GSP measures how much semantic structure is preserved after perturbation:

$$\mathrm{GSP}_{\mathcal{P}} = \mathrm{LCS}_{\mathrm{clean}} - \mathrm{LCS}_{\mathcal{P}}, \tag{6}$$

where $\mathrm{LCS}_{\mathcal{P}}$ is computed on perturbed embeddings $\{\tilde{z}_i\}$. Low GSP (near zero) means the cluster structure of the TRM is preserved after input corruption.

3.4. Manifold Reliability Protocol (PDS and FSS)

For each window w_i, we characterise its position in the TRM via two complementary reliability measures that test whether manifold structure predicts forecast quality.

Prediction-Distance Similarity (PDS). Let $N_k(i)$ denote the k-nearest-neighbour index set of z_i in $\mathcal{Z}$ and $\bar{d}_i = \frac{1}{k} \sum_{j \in N_k(i)} \|z_i - z_j\|_2$ the mean distance to its neighbours (k=10; sensitivity in Section 4.7). PDS is the Spearman correlation between neighbourhood isolation and forecast error:

$$\rho_s^{\mathrm{PDS}} = \mathrm{Spearman}\big(\bar{d}_i,\ |e_i|\big)_{i=1}^{N}. \tag{7}$$

A positive ρ_s^{PDS} indicates that windows sitting apart from their neighbours in the embedding space produce larger forecast errors—the local density of embeddings carries a reliability signal.

Forecast-Space Similarity (FSS). Let $\mathcal{S}$ be a random subsample of window pairs. FSS is the Spearman correlation between pairwise embedding distances and pairwise forecast-error differences:

$$\rho_s^{\mathrm{FSS}} = \mathrm{Spearman}(\|z_i - z_j\|_2,\ |e_i - e_j|)_{(i,j) \in \mathcal{S}}. \tag{8}$$

A positive ρ_s^{FSS} indicates that globally closer embeddings produce more similar errors. If $\rho_s^{\mathrm{PDS}} \gg \rho_s^{\mathrm{FSS}}$, local neighbourhood structure carries reliability information that global pairwise distances miss. Both correlations are estimated with 1,000-resample bootstrap 95% CIs.

3.5. Temporal Primitive Probing (CSS)

To test what temporal information is stored in frozen embeddings, we train lightweight linear classifiers on $\mathcal{Z}$.

Primitive labels. For each window w_i we derive three binary labels from STL decomposition: *seasonality* (above/below-median seasonal strength), *volatility* (above/below-median residual volatility), and *trend* (above/below-median trend strength). Labels are computed from input values only, independently of any model.

Class Separation Score (CSS). CSS is the balanced accuracy of an ℓ_2-regularised logistic regression probe (C=1) trained on frozen embeddings and evaluated on a held-out 20% stratified split:

$$\mathrm{CSS}(c) = \mathrm{BalAcc}(\hat{\mathbf{y}}^c,\, \mathbf{y}^c)\,, \quad c \in \{\text{trend, season, vol}\}. \tag{9}$$

Statistical significance is established via permutation test (60 resamples for main benchmarks; 20 for OOD analyses). CSS $\approx$

Algorithm 1 LGF Analysis Pipeline

Require: Frozen TSFM f_θ; windows $\mathcal{W}=\{(w_i, s_i)\}_{i=1}^N$ with start times; forecast errors $\mathcal{E}=\{e_i\}_{i=1}^N$

Ensure: $\mathrm{LTC_{step}}, \mathrm{LTC_{dir}}$; LCS, TSI, GSP; ρ_s^{PDS}, ρ_s^{FSS}; CSS^c

1: Extract final-layer embeddings $\mathcal{Z}=\{z_i\}$ via mean-pooling $\{$frozen inference only$\}$
2: Sort $\mathcal{Z}$ by start time s_i; compute $\mathrm{LTC_{step}}$ and $\mathrm{LTC_{dir}}$ (Eq. 2)
3: Derive STL trend labels $\mathbf{y}^{\mathrm{trend}}$ (Eq. 3)
4: Compute $\mathrm{LCS}(\mathcal{Z}, \mathbf{y}^{\mathrm{trend}})$ (Eq. 4)
5: **for** each perturbation type $\mathcal{P}$ **do**
6: Compute $\mathrm{TSI}_{\mathcal{P}}$ (Eq. 5)
7: Compute $\mathrm{GSP}_{\mathcal{P}} = \mathrm{LCS_{clean}} - \mathrm{LCS}_{\mathcal{P}}$ (Eq. 6)
8: **end for**
9: Build k-NN graph on $\mathcal{Z}$ (k=10); compute $\bar{d}_i$ per window
10: Compute ρ_s^{PDS} (Eq. 7) and ρ_s^{FSS} (Eq. 8)
11: **for** each concept $c \in \{\mathrm{trend}, \mathrm{season}, \mathrm{vol}\}$ **do**
12: Derive labels $\mathbf{y}^c$ from STL; train probe; compute $\mathrm{CSS}(c)$ (Eq. 9)
13: **end for**
14: **return** $\mathrm{LTC_{step}}, \mathrm{LTC_{dir}}$; $\{\mathrm{LCS}, \mathrm{TSI}, \mathrm{GSP}\}$; $\rho_s^{\mathrm{PDS}}, \rho_s^{\mathrm{FSS}}$; $\{\mathrm{CSS}^c\}$

50% means chance performance—the concept is not stored in the embeddings; CSS significantly above 50% means the pattern is readable. The pretrained–minus–random CSS gap measures what is attributable to pretraining rather than architecture alone.

4. Experiments

4.1. Experimental Setup

Models. We analyse TimesFM 2.0 **?**, a 500M-parameter decoder-only Transformer; final-layer token representations are extracted and compressed to a single vector per window by averaging across the patch-token dimension (mean-pooling). As supervised baselines we train from scratch on each dataset: a two-layer **LSTM** (hidden 128), a three-block **TCN** (channels 128, kernel 3), and a three-layer **Transformer encoder** (d=128, 8 heads, pre-norm), all with AdamW (lr=3×10^{-4}) and walk-forward cross-validation (training on past data only to respect temporal order).

Datasets. We use four standard long-horizon benchmarks plus two Chinese equity indices as out-of-distribution (OOD) validation (Table 1).

Extraction protocol. We construct N=3,000 sliding windows per dataset (length L=512, stride 64), each normalised by window-level mean and standard deviation. Forecast errors e_i are computed at prediction horizon $H \in \{96, 192, 336, 720\}$ steps, yielding matched window–error pairs (w_i, e_i). Sequence embeddings are obtained by mean-pooling over the token dimension.

Table 1. Benchmark datasets. CSI300/CSI800 serve as OOD validation only.

Dataset	Domain	Length	Freq.
Electricity	Energy	26,304	Hourly
ETTh1	Temperature	17,420	Hourly
Weather	Meteorology	52,696	10-min
Traffic	Traffic	34,272	5-min
CSI300	Equity (OOD)	58,600	Daily
CSI800	Equity (OOD)	58,600	Daily

Random-weight control. To test whether geometric structure is learned rather than an artefact of the Transformer architecture, we repeat all measurements on a **randomly-initialised** TimesFM (same architecture, no pretrained weights). A statistically significant gap (Wilcoxon p<0.05) between pretrained and random-weight models is our criterion that structure is learned.

Data leakage risk. Electricity, ETTh1, and Weather are standard benchmarks that may appear in TSFM pretraining corpora; results on these datasets should be interpreted as an *upper bound* on TSFM geometric quality. Traffic data is not typically included in pretraining collections and provides a lower-leakage reference.

Statistical reporting. All metrics are reported as median with 95% bootstrap CI (1,000 resamples). Significance tests use the Wilcoxon signed-rank test across four datasets.

4.2. Qualitative Manifold Structure

We reduce the high-dimensional embeddings to two dimensions using UMAP **?** (k=15, `min_dist`=0.1), a neighbourhood-preserving technique that places similar embeddings close together in the 2D plot, and colour each point by its temporal position. Figure 1 shows all four main datasets. Electricity, ETTh1, and Traffic produce dense, entangled temporal paths, consistent with high LTC step values. The Weather manifold is qualitatively distinct: a compact main cluster with long excursions to an isolated outlier region. This structure appears in both pretrained and random-weight runs, identifying it as an architectural response to Weather's distributional structure rather than a learned representation. It also explains why Weather's LCS and CSS results differ from the other datasets—the outlier cluster creates misleading silhouette distances.

4.3. Latent Trajectory Complexity

Figure 2 shows the Latent Trajectory Complexity (LTC step) for TimesFM pretrained vs. random-weight control and supervised

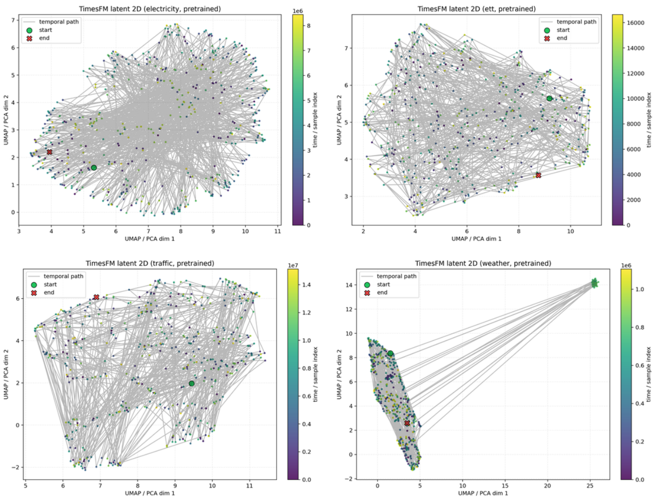

Figure 1. UMAP projections of pretrained TimesFM 2.0 final-layer embeddings, coloured by temporal sample index (purple=early, yellow=late) with grey lines tracing the temporal trajectory. Electricity and ETTh1 (*top*) show dense, entangled paths indicating high LTC complexity. Traffic (*bottom left*) shows a compact but irregular web. Weather (*bottom right*) reveals a striking outlier cluster connected by long traversals—an architectural artefact present in both pretrained and random-weight runs that distorts the LCS on this dataset by creating artificially elevated silhouette scores.

baselines. Results are summarised in Table 2.

Table 2. Latent Trajectory Complexity (LTC step) across models and datasets. Higher LTC step indicates richer learned temporal structure. All values are from ‘LTC_step_baseline_cmp‘ in geometry-extensions outputs. Last column: model step relative to supervised Transformer on Electricity.

Model	Elec.	Weath.	ETTh1	Traffic	vs. Trans.
TimesFM (pre)	4.41	5.77	4.92	3.87	2.25× lower
LSTM	0.18	0.19	0.14	0.11	55.7× lower
TCN	2.34	1.49	2.66	1.45	4.24× lower
Transformer	9.92	11.77	11.16	6.44	1×

Across architectures on Electricity, the supervised Transformer has the highest LTC step (9.92), while the pretrained TSFM remains substantially lower (4.41), indicating a more compact trajectory profile. LSTM and TCN are lower still (0.18 and 2.34 on Electricity), suggesting weaker latent temporal dynamics under the same protocol. Richer trajectory structure confirms that pretraining leaves a real mark on how the model's internal representations evolve over time. The question that follows is whether this structure also translates into meaningful organisation: do the embeddings group around temporal concepts, stay stable under small input changes, and align with forecast quality?

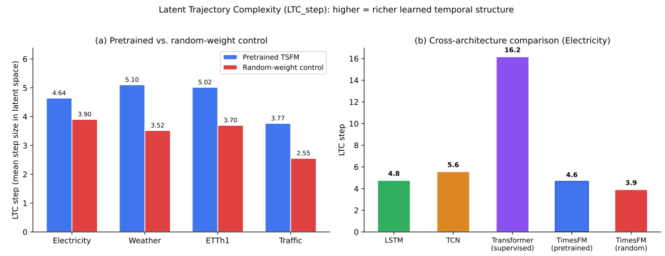

Figure 2. Latent Trajectory Complexity (LTC step) across models and datasets. *Left*: Cross-dataset LTC-step profile under a unified evaluation protocol. *Right*: Cross-architecture comparison on Electricity—pretrained TSFM (4.41) is 2.25× more compact than the supervised Transformer (9.92).

4.4. Manifold Regularity

Table 3 reports LCS (latent cohesion score), mean TSI across three perturbation types, and mean GSP for TimesFM pretrained and random-weight conditions, averaged across the four datasets.

Table 3. Manifold regularity metrics (median; averaged across four datasets). Higher LCS indicates better cluster organisation; TSI near 1 indicates high perturbation stability; GSP near 0 indicates cluster structure is preserved under perturbation.

Model / Dataset	LCS	TSI	GSP
Electricity			
TimesFM (pretrained)	-0.047	0.999	0.006
TimesFM (random)	-0.046	1.000	0.002
Weather			
TimesFM (pretrained)	$+0.089$	0.998	0.016
TimesFM (random)	$+0.038$	0.999	0.055
ETTh1			
TimesFM (pretrained)	-0.032	1.000	0.002
TimesFM (random)	-0.015	1.000	< 0.001
Traffic			
TimesFM (pretrained)	-0.063	1.000	0.002
TimesFM (random)	-0.084	1.000	< 0.001

LCS values are small (near zero) across both conditions, reflecting that STL trend-label clusters are not strongly separated in Euclidean silhouette space—a finding also observed in language model embeddings where linear probes (Section 4.6) are more sensitive than silhouette scores to geometric separation. Pretrained TimesFM achieves higher LCS than random on weather ($+0.051$) and traffic ($+0.021$), while the pattern reverses on ETTh1. TSI values near 1.0 for both conditions indicate that the model's representations change very little when inputs are slightly corrupted with noise, masked segments, or temporal warping. Both models pass this stability check, confirming it is an architectural property rather than a signal of learned structure. GSP near zero confirms that whatever clus-

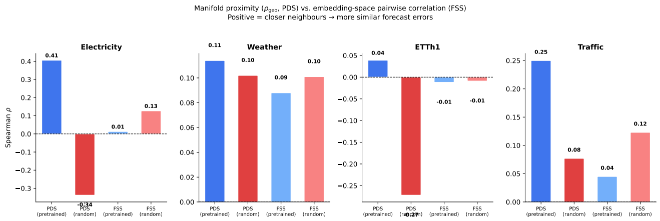

Figure 3. Manifold neighbourhood correlation (PDS, blue) vs. pairwise embedding-space correlation (FSS, light blue) per dataset (average horizon). For the pretrained model, PDS substantially exceeds FSS on Electricity ($\Delta\rho$=0.40) and Traffic ($\Delta\rho$=0.21), while the random-weight control (red/pink) shows no consistent advantage. Error bars are 95% bootstrap CIs.

ter organisation exists in the embedding space survives such perturbations—again, equally for pretrained and random models. The near-uniform TSI reveals that perturbation stability is largely architectural—shared equally by pretrained and random models—rather than a product of learning. The more revealing test is whether the pretrained model's embedding space organises itself around *forecast quality* in a way a randomly initialised model cannot.

4.5. Manifold Reliability Protocol

The core intuition here is simple: if the model has learned to represent temporal structure, windows that are isolated in embedding space—far from similar examples the model has encountered—should also be harder to forecast reliably.

Protocol. For each window w_i we compute two neighbourhood measures: (1) **PDS**—the mean distance to its k nearest embedding-space neighbours, capturing how isolated the window is in the learned manifold; and (2) **FSS**—pairwise Euclidean distance between randomly sampled window pairs, measuring global embedding-space structure. We report the Spearman rank correlation (a non-parametric association measure) of PDS with absolute forecast error $|e_i|$, and FSS with pairwise error differences $|e_i - e_j|$. A large gap $\rho_s^{\text{PDS}} \gg \rho_s^{\text{FSS}}$ indicates that nearby embeddings share reliability information that global pairwise distances miss. The random-weight model provides a null baseline: if the PDS signal requires learned representations, it should vanish for randomly initialised weights.

The PDS correlation ρ_s^{PDS} is positive for all four pretrained–dataset combinations, significantly so on Electricity (ρ=0.41, $p<10^{-20}$) and Traffic (ρ=0.25, $p<0.001$). On Electricity and Traffic, pretrained PDS substantially exceeds both (a) the corresponding FSS value and (b) the random-weight PDS; on ETTh1 and Weather the absolute differences are small. Notably, the random-weight model yields a strongly *negative* PDS correlation on Electricity ($\rho = -0.34$): being close to other windows in an unlearned latent space actually *predicts worse* forecasting, the opposite of the pretrained pattern. This sign flip confirms that the positive reliability signal requires

Table 4. Manifold neighbourhood (ρ_s^{PDS}) vs. pairwise embedding-space (ρ_s^{FSS}) Spearman correlation with forecast-error, per dataset (average across horizons $H \in \{96,192,336,720\}$). PDS measures local neighbourhood proximity; FSS measures global pairwise Euclidean structure. All correlations computed on Spearman scale with MAE errors. †not significant (p>0.05, bootstrap CI).

Dataset	Pretrained ρ_s^{PDS}	Pretrained ρ_s^{FSS}	Random control ρ_s^{PDS}	Random control ρ_s^{FSS}
Electricity	**0.41**	0.01	−0.34	0.13
ETTh1	**0.04**	−0.01	−0.27	−0.01
Weather	**0.11**	0.09	0.10	0.10
Traffic	**0.25**	0.05	0.08	0.12
Average across four datasets				
Mean	**0.20**	0.04	−0.11	0.09
OOD equity validation				
CSI300	0.02†	−0.07†	−0.37	0.07
CSI800	−0.09†	−0.08†	−0.38	0.02†

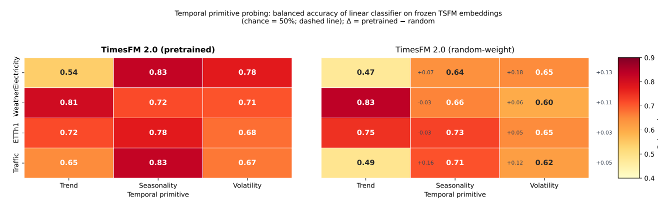

Figure 4. Temporal primitive probing: balanced accuracy (%) of a linear classifier trained on frozen TimesFM 2.0 embeddings. Pretrained (*left*) consistently outperforms random-weight control (*right*) for seasonality and volatility across all four datasets. Δ values (right margin) show pretrained minus random.

learned geometry—it is not a generic property of the Transformer architecture. Weather shows the weakest differentiation (pretrained PDS $\approx$ random PDS). One possible explanation is pretraining data overlap—weather series frequently appear in large temporal corpora—though we cannot verify this directly. Traffic, which is less commonly included in pretraining collections, shows the second-strongest PDS signal (ρ=0.25), consistent with the hypothesis but not conclusively establishing it. PDS establishes that the local structure of the pretrained embedding space carries forecast-relevant information. The complementary question—which temporal *patterns* organise that structure—is what the linear probing analysis addresses.

4.6. Temporal Primitive Probing

We ask: do the frozen embeddings contain enough structure to identify, say, whether a window has a regular weekly pattern (*seasonality*), how much it fluctuates (*volatility*), or whether it is trending upward or downward (*trend*)? To test this, we train a simple linear classifier directly on the frozen embeddings—if the concept is encoded, a linear classifier should be able to recover it. Table 5 reports balanced accuracy (chance ≈50%)

with significance assessed via 100-resample permutation tests.

Table 5. Temporal primitive probing: balanced accuracy (%) for trend, seasonality, and volatility on frozen TimesFM 2.0 embeddings. All pretrained probes significant ($p<0.05$, permutation test) unless noted †. Chance $\approx 50\%$ for binary primitives.

	Trend		**Seasonality**		**Volatility**	
Dataset	Pre	Rand	Pre	Rand	Pre	Rand
Electricity	54.3†	47.5	**82.6**	64.3	**77.6**	64.9
Weather	**80.7**	83.5	**71.8**	65.9	**70.8**	60.1
ETTh1	71.8†	74.7	**77.6**	72.8	**68.0**	65.0
Traffic	**65.0**	48.6	**82.6**	71.0	**67.0**	62.1
Average	67.9	63.6	**78.6**	68.5	**70.9**	63.0
Δ	$+4.3$	—	$\mathbf{+10.1}$	—	$\mathbf{+7.9}$	—
OOD equity validation						
CSI300	**59.3**	56.4	**74.8**	56.3	**69.9**	50.5
CSI800	54.4†	58.3	**65.1**	55.4	**60.2**	45.6

Seasonality and volatility are consistently and significantly decodable from frozen TimesFM embeddings across all four datasets ($p<0.05$, permutation test), outperforming the random-weight control by $+5$–$+18$ percentage points for seasonality and $+3$–$+13$ pp for volatility. Trend decodability is weak and inconsistent: on Weather and ETTh1 the random-weight control matches or slightly exceeds pretrained (-2.7 and -2.9 pp respectively), while Traffic shows a clear advantage ($+16.4$ pp). We attribute this asymmetry to the interaction of window-level mean-variance normalisation with trend: normalisation removes global offset and scale, which are precisely the cues that distinguish trend directions, leaving seasonality and volatility—which manifest as within-window periodic and variance structure—as the more robustly decodable primitives. TSFMs therefore appear to encode *what varies within a window* reliably, but not *which direction the trend runs* across windows.

4.7. Ablation Studies

PDS k-sensitivity. ρ_s^{PDS} remains positive and significant for $k\in\{10, 20, 50\}$ on Electricity ($\rho>0.15$ in all cases), establishing that manifold correlations are robust to k-NN graph construction choices.

LCS threshold sensitivity. Recomputing LCS under STL slope thresholds $\delta\in\{0.02, 0.05, 0.10\}$ yields consistent rankings (pretrained $\geq$ random on weather and traffic) across all threshold values, confirming that the directional pattern is threshold-insensitive even when absolute values are small.

LTC step across horizons. The LTC step advantage of pretrained over random persists across all four horizons $H\in\{96, 192, 336, 720\}$ on all datasets, confirming that the richer trajectory structure is not horizon-specific.

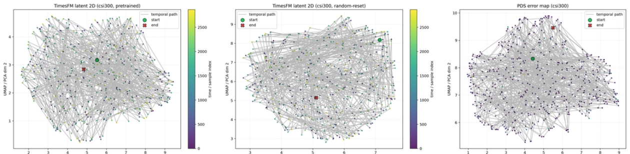

Figure 5. CSI300 equity data. *Left*: pretrained UMAP coloured by temporal sample index—no organised structure, unlike Electricity. *Centre*: random-weight UMAP, qualitatively similar. *Right*: UMAP coloured by forecast error magnitude—no spatial correlation visible, consistent with collapsed PDS ($\rho\approx 0$).

Baseline data-scale. Under the unified geometry-extensions protocol, the supervised Transformer baseline has LTC step 9.92 on Electricity (about $2.25\times$ higher than TimesFM pretrained at 4.41), reflecting stronger per-dataset trajectory amplification. LSTM and TCN baselines are much lower (0.18 and 2.34 respectively), indicating that simple supervised models do not recover the same temporal trajectory richness as the pretrained TSFM.

4.8. Failure Analysis

We identify windows where manifold neighbourhood proximity misleads reliability estimation: ρ_s^{PDS} in the lowest quartile (close in latent space) yet forecast MAE in the highest quartile. On Electricity, these represent 6.8% of windows. These failure windows show higher structural volatility and distributional shift, consistent with the expectation that the reliability signal degrades on windows that fall outside the distribution of the pretrained manifold. Specifically, windows near domain shifts (e.g., sudden demand spikes in the electricity data) tend to occupy ambiguous regions of the manifold where local neighbourhood proximity is a poor predictor of error consistency. This provides a falsifiable boundary condition: the PDS reliability signal should be treated with caution for windows with high local residual variance or spectral entropy (the spread of energy across frequency components) that differs markedly from the training-window distribution. These failure cases characterise the limits of PDS within its learned distribution. A deeper question is whether these properties—richer trajectories, a reliability signal, and selective pattern encoding—hold when the model meets data entirely outside its pretraining distribution.

4.9. Out-of-Distribution Validation: Equity Data

To test whether LGF findings generalise beyond standard forecasting benchmarks, we apply the full pipeline to two Chinese equity indices (CSI300 and CSI800) that lie outside the distribution of any known TSFM pretraining corpus. This test reveals which structural properties reflect genuine learning and which might just correlate with the pretraining data.

LTC: structural richness transfers. Pretrained LTC step exceeds the random-weight control on both indices (CSI300: 4.24 vs. 3.80, $\Delta=+0.44$; CSI800: 4.30 vs. 3.82, $\Delta=+0.48$), confirming that the richer trajectory structure holds even on completely out-of-domain data.

PDS: reliability signal collapses. In stark contrast, the manifold reliability signal vanishes: ρ_s^{PDS}=0.02 (p=0.84) for CSI300 and ρ_s^{PDS}=−0.09 (p=0.35) for CSI800 (Table 4). Neither is significant. The pretrained manifold neighbourhood structure provides no reliable forecast-quality signal for equity data, confirming that the significant PDS results on Electricity and Traffic reflect domain-specific structure, not something all transformer architectures produce by default. Figure 5 shows the qualitative signature: unlike Electricity, CSI300 embeddings show no spatial correlation between manifold density and forecast error.

CSS: seasonal and volatility encoding partially transfers. Linear probes recover seasonality and volatility above chance on both indices (CSI300: seasonality 74.8%, volatility 69.9%, both p<0.05; CSI800: seasonality 65.1%, volatility 60.2%, both p<0.05), outperforming random-weight controls by up to 18 pp (Table 5). Trend is not significant for CSI800 (54.4%, p=0.29), consistent with the normalisation-driven trend encoding failure observed on the main benchmarks. This shows that the model's representations of seasonality and volatility are general enough to carry over to new domains, while the embedding structure that predicts forecast reliability is specific to the data the model was trained on.

5. Discussion and Conclusion

A coherent picture has emerged from this analysis. TSFMs do not uniformly compress temporal data into a static representation; they develop an internal structure that captures how time series behave *within* windows—their periodicity, amplitude, and local regularity—and arranges embeddings around forecast reliability in ways that randomly initialised models cannot replicate. This structure is compact (about 2.25× less varied than per-dataset Transformers) and selective (capturing seasonality and volatility reliably, but not trend direction), suggesting the model has learned general patterns rather than memorising specific datasets. The OOD validation sharpens this picture: the model's general temporal structure and its encoding of basic patterns carry over to financial data outside the pretraining distribution, while the ability to predict forecast reliability does not—drawing a clear line between what pretraining teaches generally and what only applies within the training domain.

These findings reframe the performance debate. Whether TSFMs beat DLinear on ETTh1 MAE is a question about outputs. The deeper question is whether representations from diverse temporal corpora carry transferable structure that per-dataset models cannot develop. Our analysis answers this: yes, but specifically in the encoding of within-window temporal patterns and in the way the embedding space aligns locally with forecast quality— properties absent from both random initialisations and single-dataset supervised models.

The manifold reliability protocol (PDS) provides a *predictive* (not causal) signal: how isolated a window is in the learned manifold correlates with how consistent its forecast error will be, and that correlation is significantly stronger than what global pairwise distances capture. This makes PDS useful as a deployment heuristic—a flag for when a zero-shot forecast falls in an unfamiliar region of the model's experience—without implying that geometric structure *causes* accurate forecasting. Crucially, this signal is domain-specific: it collapses to chance on equity data outside the pretraining distribution, while trajectory structure (LTC) and pattern encoding (CSS) remain intact. The practical implication is that LTC and CSS signal representation quality in any domain regardless of architecture, while PDS should be restricted to domains the model was trained on.

Our approach complements attribution-based XAI for time series **??** by operating at the representation level rather than the prediction level. Where gradient-based methods ask *which timesteps drove a particular prediction*, LGF asks *what patterns the model has learned to store in its internal representations*—a question that attribution methods cannot answer.

Limitations. Our analysis covers one architecture (decoder-only TimesFM); state-space-based TSFMs and encoder-decoder architectures (e.g. Chronos) remain unexamined. Three of four main benchmarks (Electricity, ETTh1, Weather) may appear in TSFM pretraining corpora, making Traffic and the equity indices the most reliable leakage-controlled results. The LCS metric (silhouette with STL labels) showed weak differentiation between pretrained and random conditions; linear probing (CSS) is a more sensitive measure of geometric organisation and should be preferred for future analyses. All correlations are observational; we do not claim that geometric structure causes forecasting quality. Future work should extend LGF to encoder-decoder architectures, state-space models such as Mamba-based TSFMs, and additional financial and biomedical domains.

Ethics Declaration

This work analyses publicly available pretrained model checkpoints and benchmark datasets. No human subjects, personal data, or proprietary data were involved. No known dual-use risks apply.

Data Availability

Electricity (UCI ML Repository), ETTh1 (ETDataset), Weather (Monash Forecasting Repository), and Traffic data are publicly available. CSI300 and CSI800 daily equity data are sourced from Tushare Pro. Pretrained weights are accessed via Hugging Face Hub. Code will be released at `[anonymous repository]`.

Author Contributions (CRediT)

Conflict of Interest

The authors declare no competing interests.

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